Qi Yang, Ph.D.

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Research interests

Tensor-network methods for quantum many-body physics: long-range interactions and criticality, iMPS and iPEPS, frustrated magnets and quantum spin liquids, numerical linear algebra, automatic differentiation, and scalable scientific computing.

Appointments and education

Sep 2024 - present
Postdoctoral researcher
University of Amsterdam, Institute for Theoretical Physics. Supervisor: Philippe Corboz.

Sep 2019 - Jun 2024
Ph.D. in Theoretical Physics
Institute of Physics, Chinese Academy of Sciences. Advisor: Lei Wang. Thesis: Construction and contraction of two-dimensional tensor networks.

Sep 2015 - Jun 2019
B.Sc. in Theoretical Physics
University of Science and Technology of China. Advisor: Youjin Deng. GPA: 3.79/4.3.

Supervision and mentoring

2024 - present
Yining Zhang, Ph.D. co-supervision
University of Amsterdam; jointly with Philippe Corboz.

2025 - present
Willem Herter, M.Sc. co-supervision
University of Amsterdam; jointly with Philippe Corboz.

Research funding and computing

2025 - 2026
Principal Investigator, EuroHPC Development Access
Awarded 14,000 NVIDIA H100 GPU hours for large-scale tensor-network simulations.

Open-source software

Personal projects and contributions to collaborative software.

QRCTM - Reference implementation of QR-based CTMRG for efficient iPEPS contractions.

omeinsum - PyTorch tensor contractions with slicing, checkpointing, path optimization, and multi-GPU dispatch.

jaxctmrg - Pedagogical CTMRG implementation with automatic differentiation in JAX.

Gaussian-fPEPS - Tools for translating Gaussian fermionic PEPS into generic fPEPS.

varbench - Collaborative benchmarks of variational methods for quantum many-body problems.

TeneT.jl - Julia library for two-dimensional infinite tensor networks.

Publications

7 journal articles and 1 preprint. † Equal contribution.

Journal articles

1. Xing-Yu Zhang†, **Qi Yang**†, Philippe Corboz, Jutho Haegeman, Yuchi He.

Topological and Trivial Valence-Bond Orders in Higher-Spin Kitaev Models

Physical Review Letters 137, 176502 (2026). DOI: 10.1103/7q31-xxsd

2. Yining Zhang†, **Qi Yang**†, Philippe Corboz.

Accelerating two-dimensional tensor network contractions using QR decompositions

Physical Review B 113, L201106 (2026). DOI: 10.1103/ptls-kr9z

3. Xing-Yu Zhang†, **Qi Yang**†, Philippe Corboz, Jutho Haegeman, Wei Tang.

Accelerating two-dimensional tensor network optimization by preconditioning

Physical Review B 113, 125111 (2026). DOI: 10.1103/h396-yc28

4. Qi Yang, Philippe Corboz.

Efficient iPEPS simulation on the honeycomb lattice via QR-based corner transfer matrix renormalization group

Physical Review B 113, 085109 (2026). DOI: 10.1103/9gmp-byx8

5. Yantao Wu, Zhehao Dai, Sajant Anand, Sheng-Hsuan Lin, **Qi Yang**, Lei Wang, Frank Pollmann, Michael P. Zaletel.

Alternating and Gaussian Fermionic Isometric Tensor Network States

PRX Quantum 6, 040324 (2025). DOI: 10.1103/8ypw-c8t4

6. Dian Wu, Riccardo Rossi, Filippo Vicentini, *et al.* (including **Qi Yang**).

Variational benchmarks for quantum many-body problems

Science 386, 296-301 (2024). DOI: 10.1126/science.adg9774

7. Qi Yang, Xing-Yu Zhang, Hai-Jun Liao, Hong-Hao Tu, Lei Wang.

Projected d-wave superconducting state: A fermionic projected entangled pair state study

Physical Review B 107, 125128 (2023). DOI: 10.1103/PhysRevB.107.125128

Preprint

Qi Yang.

Transfer-matrix functions for algebraically decaying interactions in variational infinite matrix product states

arXiv:2606.20522 (2026)

Talks and academic activities

Invited talks and lectures

2026-07-27 **How to talk to the Wish Machine?**
Harnessing Quantum 2026, online invited lecture (40 min).
Program

May 2026 **iPEPS Simulations of 2D Strongly Correlated Systems in the Era of AD and GPUs**
Invited seminar, Hangzhou, China.

Apr 2026 **Scalable iPEPS Simulations in the Era of AD and GPUs**
Invited seminar, Hefei, China.

Apr 2026 **When Corners Meet Edges: Emergent Algebra in 2D TN Boundaries**
Invited seminar, Beijing, China.

2026-04-12 **QRCTM: scalable iPEPS simulation for frustrated quantum magnets**
Invited presentation, 14th Workshop on Quantum Many-Body Computation (QMBC-2026), Beijing, China.
Slides

2026-01-27 **Scalable iPEPS Simulations in the Era of AD and GPUs**
Invited seminar talk, Ludwig Maximilian University of Munich, Germany.
Slides

Workshop talk

2026-01-14 **Accelerating two-dimensional tensor network contractions using QR-decompositions**
Fixed-Point Optimization in Tensor Networks for Quantum Matter, ETH Zurich, Switzerland.
Slides

Contributed talk

2025-10-10 **Exploring the Ground States of Higher-Spin Kitaev Models via QR-Accelerated iPEPS simulation**
A Panorama of Tensor Networks, Krakow, Poland.
Slides

Awards

2022 - Outstanding Scholarship, IOP, Excellent Prize (top 10%)
2019 - Outstanding Students Scholarship, IOP, B-class
2018 - Competition of Physical Experiments, USTC, Premium Prize
2017 - Academic Scholarship, USTC, Silver Prize
2016 - Academic Scholarship, USTC, Bronze Prize
2014 - Chinese Physics Olympiad, Anhui, First Prize; Chinese Mathematics Olympiad, Anhui, Second Prize